

Exercise 1

Consider the two-stage amplifier in Fig. 1. The circuit parameters, at $T = 25^\circ\text{C}$, are the following:

$V_{CC} = 12\text{ V}$, $R_S = 33\text{ k}\Omega$, $R_{B1} = 120\text{ k}\Omega$, $R_{B2} = 12\text{ k}\Omega$, $R_C = 22\text{ k}\Omega$, $R_{E1} = 1\text{ k}\Omega$; $R_{E2} = 12\text{ k}\Omega$, $R_L = 10\text{ k}\Omega$, $R_F = 150\text{ k}\Omega$.

Q_1 : $V_{BE} = 0.7\text{ V}$; $\beta_F = 100$; $\beta_0 = 100$; $r_0 = +\infty$;

Q_2 : $V_{BE} = -0.7\text{ V}$; $\beta_F = 100$; $\beta_0 = 100$; $r_0 = +\infty$;

$V_T = 25\text{ mV}$.

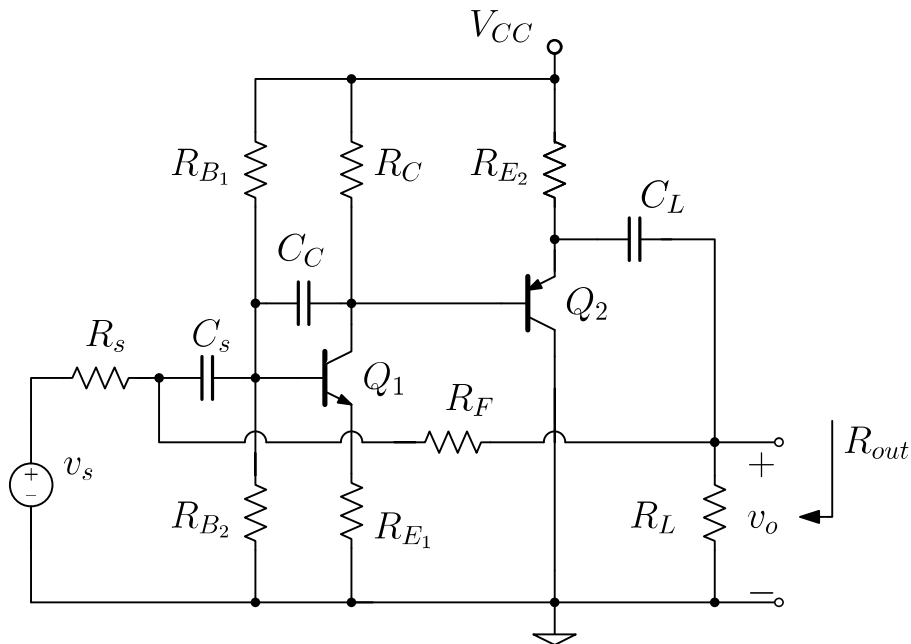


Fig. 1 – Two stage amplifier with feedback.

Exercise 2

Consider the operational amplifier configuration shown in Fig. 2. The circuit parameters are the following:

$R_1 = 1 \text{ k}\Omega$; $R_2 = 100 \text{ k}\Omega$; $C_1 = 1 \text{ nF}$;

$$A(s) = \frac{10^5}{1 + \frac{s}{10^2}} \text{ [V/V]}.$$

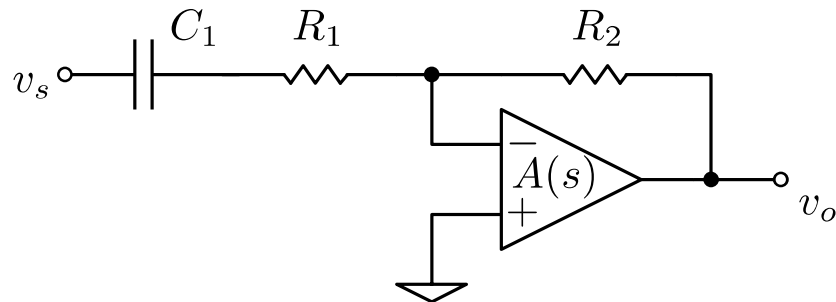


Fig. 2 – High-pass inverting configuration of an operational amplifier.